

Prompt Engineering: Week 1

Introduction to Prompt Engineering

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Today: What We'll Explore

- What is a "prompt"? A friendly, everyday definition
- Why prompts matter for everyone (not just programmers)
- Core terms: Prompt, LLM, Generative AI, Context
- Examples across fields: creative writing, law, marketing, engineering, education, medicine
- In-class activity: vague vs detailed prompts
- Common pitfalls and quick wins
- Assignment for next week

What is a "Prompt"?

- A **prompt** is how you talk to an AI system. It's the instruction, question, or example you give it.
- Everyday analogy: asking Google vs asking a friend. The way you ask changes the answer you get.
- Think of a prompt as a **brief** you give to a creative assistant.
- Good prompts provide: **purpose**, **context**, and **constraints**.

Why Prompting Matters in 2025

- AI is a general-purpose tool: writing, summarizing, analyzing, creating.
- Prompting is a new form of **digital literacy**.
- Every discipline benefits:
 - Medicine: summarizing patient histories, drafting patient education.
 - Law: drafting clauses, summarizing cases, spotting issues.
 - Business: market research, pitch drafts, customer support.
 - Engineering: code generation, documentation, debugging help.
 - Design: brainstorming, mood boards, content ideas.
- With better prompting, you get **clearer, safer, more useful** outputs.

Key Terms (Plain English)

- **Prompt:** The input message you give the AI.
- **LLM (Large Language Model):** A system trained on lots of text to predict the next word.
- **Generative AI:** AI that can create new content (text, images, code) based on patterns it learned.
- **Context:** The information the AI considers for a response, including your conversation history.

Example: Creative Writing

Vague

Write a story about a journey.

Better

You are a literary author. Write a 500-word short story in the style of magical realism about a paramedic in Istanbul who discovers time pauses whenever it rains. Use vivid sensory detail and a hopeful tone.

- Why it's better: role, style, length, setting, tone, constraint.

Example: Legal Drafting

Vague

Draft a contract.

Better

You are a contract lawyer. Draft a one-page independent contractor agreement for a freelance graphic designer in Turkey. Include scope of work, payment schedule (milestones), IP ownership (work-for-hire), confidentiality, and termination clauses. Use clear, plain language.

- Why it's better: role, jurisdiction, length, required clauses, language.

Example: Marketing

Vague

Create a slogan for a coffee shop.

Better

You are a brand strategist. Generate 10 short, punchy slogans for a student-friendly specialty coffee shop near a university in Ankara. Emphasize affordability, community, and late-night study vibes. Return as a numbered list.

- Why it's better: role, audience, location, values, format.

Example: Business

Vague

Write a business plan.

Better

You are a startup advisor. Draft a one-page business brief for a campus coffee cart at Ankara Medipol University. Include sections: problem, solution, target customers, value proposition, pricing, go-to-market (2 channels), basic cost estimate, and key risks with mitigations. Use bullet points.

- Why it's better: scope and structure, audience, length, concrete sections, risk awareness.

Example: Data Analysis

Vague

Analyze this sales data.

Better

You are a data analyst. Given a CSV with columns date, product, region, and revenue, compute monthly revenue by region for the last 12 months, identify any months more than 2 standard deviations from the 12-month regional mean, and return: (1) a 4-row table (one per region) with monthly totals, (2) a brief 80–120 word summary of anomalies, and (3) recommendations in 3 bullets.

- Why it's better: clear inputs, time window, statistical rule, deliverables, and output format.

Example: Engineering

Vague

Write Python code to process data.

Better

You are a software engineer. Write a Python function that reads a CSV file with columns `time` and `voltage`; compute a rolling average over a 5-second window; handle missing values by forward fill; include a docstring and type hints; return a pandas DataFrame; and show a short usage example.

- Why it's better: role, file format, library, requirements, error handling, output spec, example.

Example: Education

Vague

Explain photosynthesis.

Better

You are a high-school biology teacher. Explain photosynthesis for 9th graders in simple language using exactly 3 bullet points and one everyday analogy. Keep it under 120 words. End with two multiple-choice quiz questions (A–D) and the answer key.

- Why it's better: role, audience level, structure, length, checks for understanding.

Example: Medicine

Vague

Summarize this patient case.

Better

You are a clinical assistant. Given a patient note, produce a concise summary with sections: HPI, PMH, meds, allergies, vitals, key labs/imaging, assessment, and suggested next steps. Use neutral, non-diagnostic language; do not fabricate missing details; add a one-line disclaimer that this is not medical advice.

- Why it's better: role, structured output, safety constraints, scope limits, disclaimer.

- **Clarity:** Make the goal explicit. What do you want?
- **Specificity:** Provide details: role, audience, length, style, format.
- **Context:** Give background, constraints, and examples.
- **Format:** Ask for structured output (bullets, table, JSON) when useful.
- **Tone:** Specify formal, friendly, persuasive, etc.

Common Mistakes and Fixes

- Too vague → add role, audience, and constraints
- Asking for everything at once → break into steps
- No examples → add 1–2 samples to guide style
- No format → request bullets or a template
- Ignoring limitations → ask for sources or confidence notes when needed

Activity: Vague vs Detailed Prompt

How we'll run it

- Pair up. Pick one task (e.g., summarize a news article, write an email, explain a concept).
- Write a **vague** prompt and run it.
- Improve it with role, audience, context, constraints, and format. Run again.
- Compare the outputs. What improved? What surprised you?

Debrief

Share one improvement that made the biggest difference.

Discussion Questions

- Where in your field could prompting save time or improve quality?
- What risks or ethical concerns should we watch out for?
- How might you evaluate whether an AI output is "good enough" for your use?

Key Takeaways

- Prompts are briefs: goal, context, constraints.
- Better prompts **consistently** lead to better outputs.
- Anyone can learn this skill with practice and iteration.
- Start simple, then refine with feedback from the output.

Assignment for Next Week

Design 3 Prompts for the Same Task

Task suggestion: Summarize a news article or explain a concept from your field.

- Prompt 1: your first attempt (vague is fine)
- Prompt 2: refined with role, audience, constraints
- Prompt 3: add examples or a specific output format

Submit the 3 prompts and the 3 outputs as a single PDF with a short reflection (150–250 words) on what changed and why.

Assignment Rubric (100%)

- **Clarity of task and prompts (25%):** Goals are explicit; instructions are unambiguous.
- **Use of prompting levers (30%):** Role, audience, context, constraints, and format are used effectively.
- **Iteration quality and reflection (25%):** Clear progression from Prompt 1 to 3; reflection explains what improved and why.
- **Professionalism and formatting (10%):** Clean structure; readable; outputs and prompts labeled.
- **Originality and appropriateness (10%):** Creative yet suitable for the chosen task and audience.

Thank You!

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Questions?